

Black Belt 6Sigma

Linear Regression

Stijn De Vuyst

Introduction to empirical models

Table 11-1 Oxygen and Hydrocarbon Levels

Observation Number	Hydrocarbon Level x (%)	Purity y (%)
1	0.99	90.01
2	1.02	89.05
3	1.15	91.43
4	1.29	93.74
5	1.46	96.73
6	1.36	94.45
7	0.87	87.59
8	1.23	91.77
9	1.55	99.42
10	1.40	93.65
11	1.19	93.54
12	1.15	92.52
13	0.98	90.56
14	1.01	89.54
15	1.11	89.85
16	1.20	90.39
17	1.26	93.25
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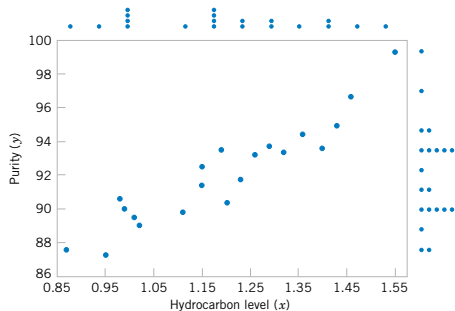


Figure 11-1 Scatter diagram of oxygen purity versus hydrocarbon level from Table 11-1.

Introduction to empirical models

Explaining the data by a model

- ▶ The data set is **univariate**: only *one* response variable y
in case the response is a vector $\mathbf{y}$, the data is called **multivariate**
- ▶ The other variables in the data set are the x -variables
 $p-1$ x -variables $x_1, \dots, x_{p-1}$ constitute the **x -domain**
→ the i th data point is $(x_{1,i}, \dots, x_{p-1,i}; y_i)$, $1 \leq i \leq n$
- ▶ In order to explain a particular **relationship** between the x -variables and their response, we may impose the following **parametric** stochastic **model**:

The responses y_i of the data points are realisations of the random variables

$$Y = f(x_1, \dots, x_{p-1}) + \varepsilon$$

where $f(\cdot)$ is a **deterministic** component
and $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ is a **stochastic error term**

Introduction to empirical models

Linear regression models

- ▶ If $f(\cdot)$ is a **linear** combination of the x -variables with an intercept, we call it

a **multiple linear regression** model

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_{p-1} x_{p-1} + \varepsilon$$

The p terms are called the **regressors** of the model and the coefficients $\beta_0, \beta_1, \dots, \beta_{p-1}$ are the **regression coefficients**

- ▶ If there is only one x -variable (i.e. $p = 2$), the model becomes

a **simple linear regression** model

$$Y = \beta_0 + \beta_1 x + \varepsilon$$

with β_0 the **intercept** and β_1 the **slope**

Introduction to empirical models

A simple linear regression model

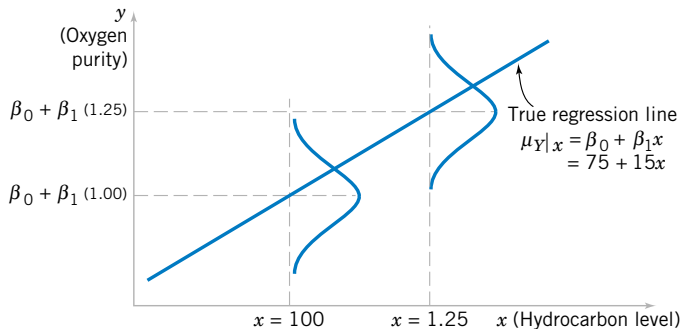


Figure 11-2 The distribution of Y for a given value of x for the oxygen purity-hydrocarbon data.

Introduction to empirical models

A multiple regression model:

$$E[Y|x_1, x_2] = 50 + 10x_1 + 7x_2$$

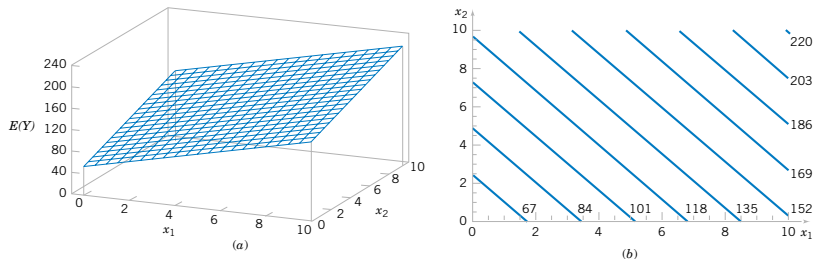


Figure 12-1 (a) The regression plane for the model $E(Y) = 50 + 10x_1 + 7x_2$. (b) The contour plot.

The regressors may also contain variables that are functions of other x -variables, e.g. on the next slide:

left: $x_3 = x_1x_2 \rightarrow$ model $E[Y] = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_1x_2$

right: $x_3 = x_1^2$, $x_4 = x_2^2$, $x_5 = x_1x_2$

$\rightarrow$ model $E[Y] = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_1^2 + \beta_4x_2^2 + \beta_5x_1x_2$

Introduction to empirical models

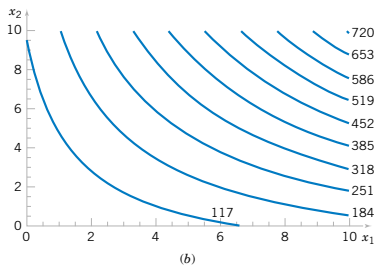
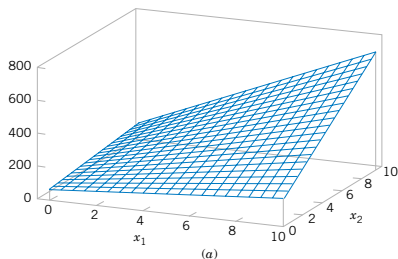


Figure 12-2 (a) Three-dimensional plot of the regression model $E(Y) = 50 + 10x_1 + 7x_2 + 5x_1x_2$. (b) The contour plot.

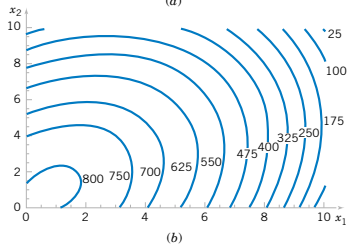
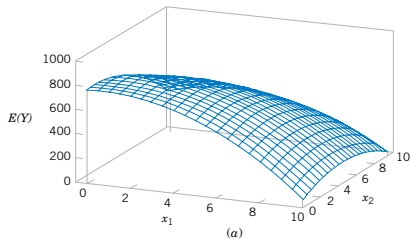
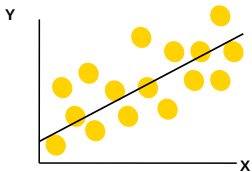


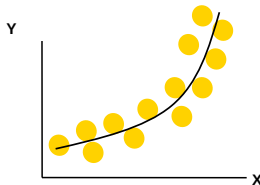
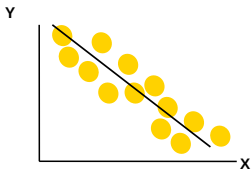
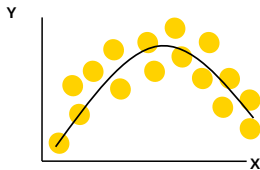
Figure 12-3 (a) Three-dimensional plot of the regression model $E(Y) = 800 + 10x_1 + 7x_2 - 8.5x_1^2 - 5x_2^2 + 4x_1x_2$. (b) The contour plot.

WELK VERBAND TUSSEN VARIABLE X EN RESPONS Y ?

lineair verband

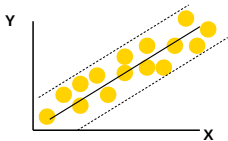


niet-lineaire relatie

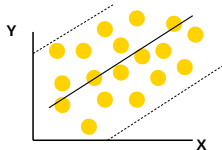


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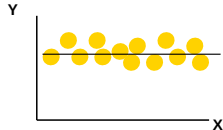
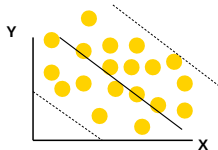
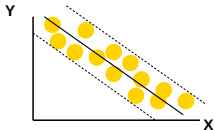
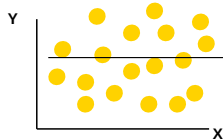
Sterke relatie



Zwakke relatie

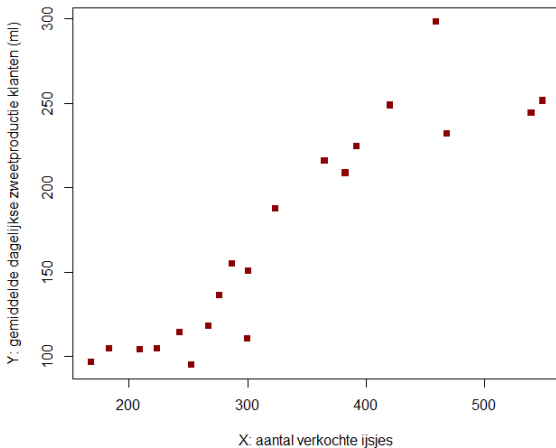


Geén relatie



WELK VERBAND TUSSEN VARIABLE X EN RESPONS Y ?

20 dagen aan zee: verband X en Y



WELK VERBAND TUSSEN VARIABLE X EN RESPONS Y ?

Belangrijk:

- Let op met “snelle” interpretatie en conclusies:

correlatie $\neq$ causaliteit

“spurious” of “confounding” relatie :

X : roomijsverkoop op een dag

Y : gemiddelde zweetproductie klanten op een dag

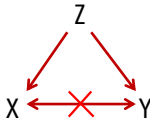
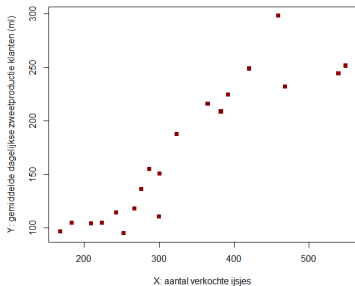
lijken verband te houden met elkaar, maar eigenlijk:

Z : dagtemperatuur

is hier de verborgen factor

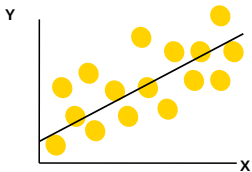
- “Resultaat” moet ook praktische betekenis hebben

20 dagen aan zee: verband X en Y

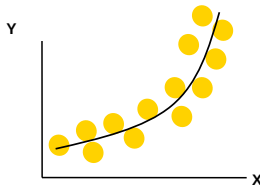
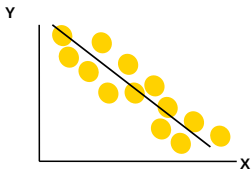
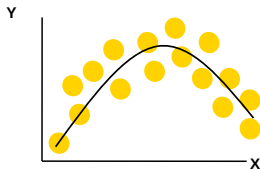


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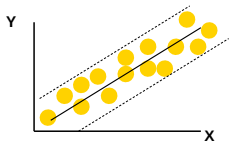


niet-lineaire relatie

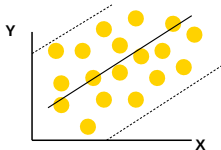


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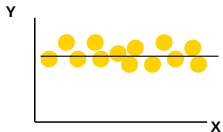
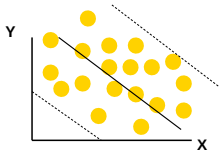
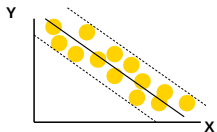
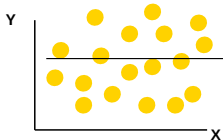
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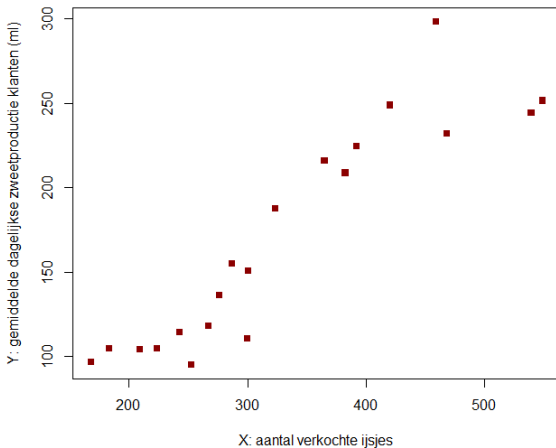


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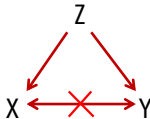
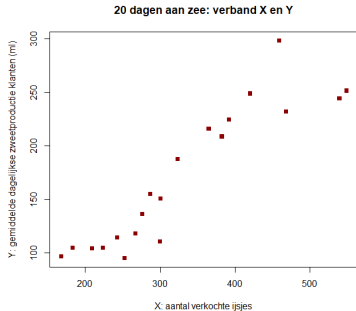
Y : gemiddelde zweetproductie klanten op een dag

lijken verband te houden met elkaar, maar eigenlijk:

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is hier de verborgen factor

- “Resultaat” moet ook praktische betekenis hebben



Simple linear regression

Assumed model

The model $Y = \beta_0 + \beta_1 x + \varepsilon$ is assumed

- Imagine an unobserved measurement (x, Y) at x , then the mean and variance of the response is:

$$\mu_{Y|x} = E[Y|x] = E[\beta_0 + \beta_1 x + \varepsilon] = E[\beta_0 + \beta_1 x] + E[\varepsilon] = \beta_0 + \beta_1 x$$

$$\text{Var}[Y|x] = \text{Var}[\beta_0 + \beta_1 x + \varepsilon] = \text{Var}[\beta_0 + \beta_1 x] + \text{Var}[\varepsilon] = 0 + \sigma^2 = \sigma^2$$

since the error ε is $N(0, \sigma^2)$ and the rest is deterministic

- This implies that at each x , the response is $Y|x \sim N(\mu_{Y|x}, \sigma^2)$

→ The variability of $Y|x$ is the same for all x !

this is called **homoscedasticity**, as opposed to **heteroscedasticity** if the variance is *not* the same for all x

Least-squares estimation

The data

Suppose we have n unobserved data points $(x_1, Y_1), \dots, (x_n, Y_n)$

Fitting the data to the model

Question:

Which linear relation $\beta_0 + \beta_1 x$ fits the data most closely?

“most closely” $\rightarrow$ smallest possible sum of squared deviations

$$L = \sum_{i=1}^n \varepsilon_i^2$$

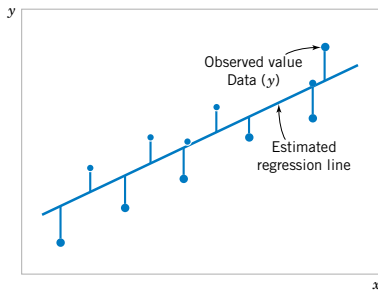


Figure 11-3 Deviations of the data from the estimated regression model.

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{\beta_0, \beta_1} L = \arg \min_{\beta_0, \beta_1} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2$$

Least-squares estimation

$$\blacktriangleright \frac{\partial L}{\partial \beta_0} = -2 \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i) = 0 \quad \longrightarrow \beta_0 = \bar{Y} - \beta_1 \bar{x}$$

$$\underbrace{\sum_i Y_i}_{n\bar{Y}} - n\beta_0 - \beta_1 \underbrace{\sum_i x_i}_{n\bar{x}} = 0$$

$$\blacktriangleright \frac{\partial L}{\partial \beta_1} = -2 \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i) x_i = 0 \quad \longrightarrow \beta_1 = \frac{S_{xY}}{S_{xx}}$$

$$\sum_i Y_i x_i - \underbrace{\beta_0}_{\bar{Y} - \beta_1 \bar{x}} n\bar{x} - \beta_1 \sum_i x_i^2 = \underbrace{\sum_i Y_i x_i - n\bar{x}\bar{Y}}_{S_{xY}} - \beta_1 \underbrace{(\sum_i x_i^2 - n\bar{x}^2)}_{S_{xx}} = 0$$

with $\bar{x} = \frac{1}{n} \sum_i x_i$, $\bar{Y} = \frac{1}{n} \sum_i Y_i$,

$$S_{xx} = \sum_i (x_i - \bar{x})^2 \quad \text{and} \quad S_{xY} = \sum_i (x_i - \bar{x})(Y_i - \bar{Y})$$

Normal equations

The **fitted intercept and slope** in the simple linear regression model are

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x} \quad (\text{NE1}) \qquad \hat{\beta}_1 = \frac{S_{xY}}{S_{xx}} \quad (\text{NE2})$$

Least-squares estimation

The fitted regression line

- The solutions of the normal equations are used as **estimates** of the true coefficients β_0 and β_1

$\mu_{Y|x} = \beta_0 + \beta_1 x \longrightarrow$ **true** (population) linear model or line

$\hat{Y}(x) = \hat{\beta}_0 + \hat{\beta}_1 x \longrightarrow$ fitted **regression line**

These are two different lines!

- Based on the available data, the regression line is our best estimate of the (unknown) true line and we use it to provide **predictions**:

at x , our best prediction/estimate of the response is $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x$

- We use the notation $\hat{\beta}_0, \hat{\beta}_1$ for both **observed** responses (x_i, y_i) and for unobserved **responses** (x_i, Y_i) , $1 \leq i \leq n$.

It should be clear from context which is meant.

Simple linear regression

Example: oxygen purity vs. hydrocarbon level

Table 11-1 Oxygen and Hydrocarbon Levels

Observation Number	Hydrocarbon Level x (%)	Purity y (%)
1	0.99	90.01
2	1.02	89.05
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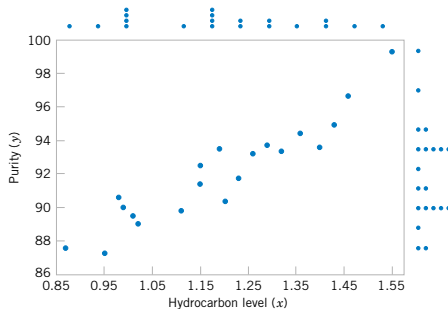


Figure 11-1 Scatter diagram of oxygen purity versus hydrocarbon level from Table 11-1.

Simple linear regression

Example: oxygen purity vs. hydrocarbon level

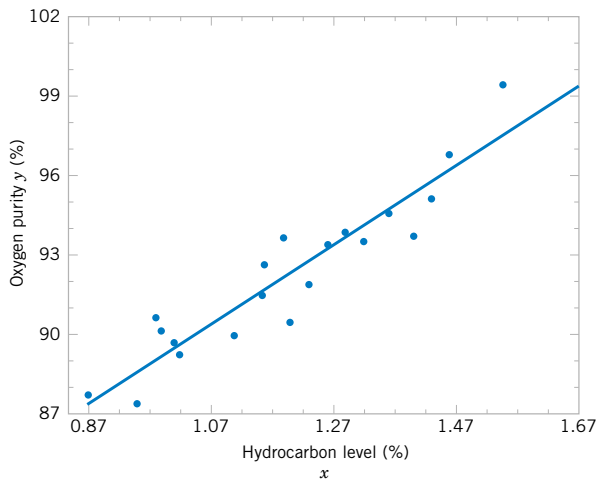


Figure 11-4 Scatter plot of oxygen purity y versus hydrocarbon level x and regression model $\hat{y} = 74.20 + 14.97x$.

Simple linear regression

Example: oxygen purity vs. hydrocarbon level

Table 11-2 Minitab Output for the Oxygen Purity Data in Example 11-1

Regression Analysis

The regression equation is

Purity = 74.3 + 14.9 HC Level

Predictor	Coef	SE Coef	T	P
Constant	74.283 $\leftarrow \hat{\beta}_0$	1.593	46.62	0.000
HC Level	14.947 $\leftarrow \hat{\beta}_1$	1.317	11.35	0.000

S = 1.087

R-Sq = 87.7%

R-Sq (adj) = 87.1%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	152.13	152.13	128.86	0.000
Residual Error	18	21.25 $\leftarrow SS_E$	1.18 $\leftarrow \hat{\sigma}^2$		
Total	19	173.38			

Predicted Values for New Observations

New Obs	Fit	SE Fit	95.0% CI	95.0% PI
1	89.231	0.354	(88.486, 89.975)	(86.830, 91.632)

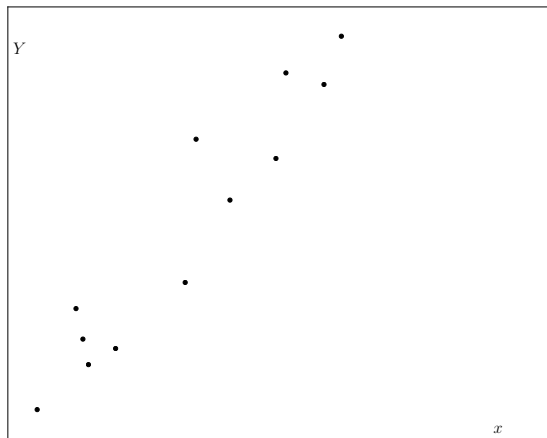
Values of Predictors for New Observations

New Obs	HC Level
1	1.00

Analysis of Variance

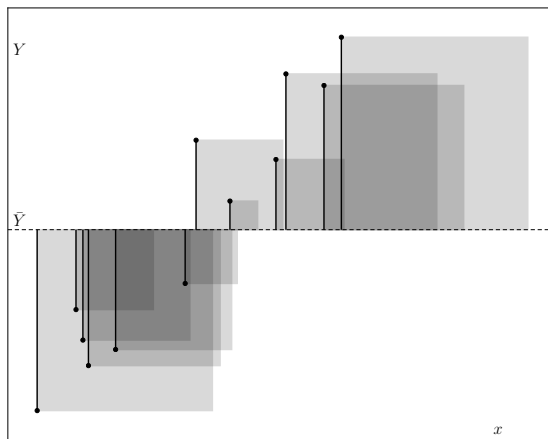
The regression model **explains variability** in the data, in the following sense . . .

We quantify “**variability**” by **sums of squares** (SS):
squared deviations from some particular mean, summed over all data points



Analysis of Variance

- ▶ **Without model**, the dependence of the data on x is neglected
→ then our best prediction for a new measurement is simply the data mean $\bar{Y}$



The total SS is

$$SS_{\text{Total}} = \sum_{i=1}^n (Y_i - \bar{Y})^2$$

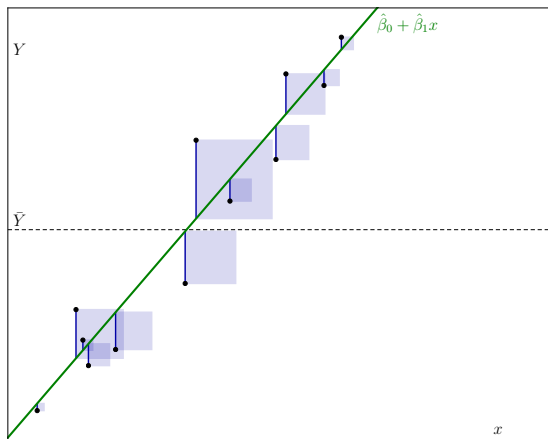
deviation of the data from
their mean $\bar{Y}$

corresponds to 'trivial' model
 $Y = \beta_0 + \varepsilon$, with $\hat{\beta}_0 = \bar{Y}$

Analysis of Variance

- ▶ With the model, the dependence of the data on x is exploited

→ our best prediction for a new measurement at x is $\hat{Y}(x) = \hat{\beta}_0 + \hat{\beta}_1 x$



The residual SS is

$$SS_{\text{Error}} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

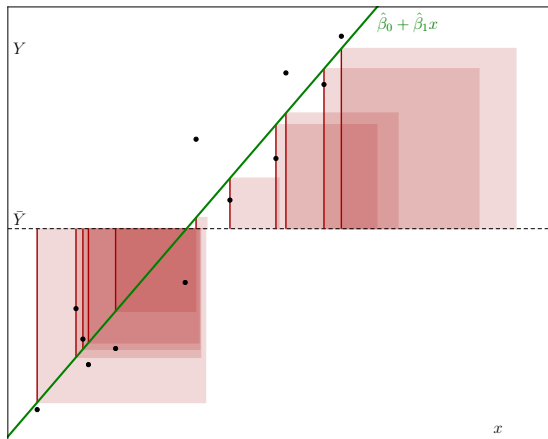
deviation of the data from
the regression line

!! This SS is a sum of squared
residuals, not errors.

Nevertheless, to comply with literature, we call it SS_{Error} or SS_E

Analysis of Variance

- The variability of **the model itself** can be seen as 'explained' variability
→ how much is using the model better than not using the model?



The **regression SS** is

$$SS_{\text{Regression}} = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$$

deviation of the regression
line from the data mean

Analysis of Variance

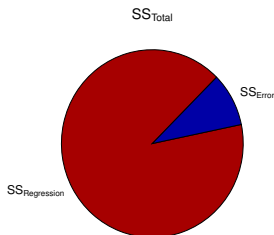
Obviously, $Y_i - \bar{Y} = \hat{Y}_i - \bar{Y} + Y_i - \hat{Y}_i$ for all i , but a remarkable property is also:

Theorem

Because of the least-squares fitting (the normal equations),

$$SS_{\text{Total}} = SS_{\text{Regression}} + SS_{\text{Error}}$$

i.e. the total variability can be partitioned in **explained** and **unexplained** variability



$$\begin{aligned} SS_T &= \sum_i (Y_i - \bar{Y})^2 = \sum_i (Y_i - \hat{Y}_i + \hat{Y}_i - \bar{Y})^2 \\ &= \underbrace{\sum_i (Y_i - \hat{Y}_i)^2}_{SS_E} + \underbrace{\sum_i (\hat{Y}_i - \bar{Y})^2}_{SS_R} + \underbrace{2 \sum_i (Y_i - \hat{Y}_i)(\hat{Y}_i - \bar{Y})}_{\substack{=0 \\ \text{by NE1/NE2}}} \end{aligned}$$

The **coefficient of determination** R^2 is defined as $R^2 = \frac{SS_R}{SS_T} = 1 - \frac{SS_E}{SS_T}$

Analysis of Variance

We can prove that:

If the **true model** for the data is indeed $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, $\varepsilon_i \sim \mathbf{N}(0, \sigma^2)$ then

SS_T has $n - 1$ degrees of freedom (df) and $E[SS_T] = \beta_1^2 S_{xx} + (n - 1)\sigma^2$

$$SS_R \quad 1 \quad E[SS_R] = \beta_1^2 S_{xx} + \sigma^2$$

$$SS_E \quad n - 2 \quad E[SS_E] = (n - 2)\sigma^2$$

$$\frac{SS_E}{\sigma^2} \sim \text{Chi}_{n-2}^2 \quad SS_R \text{ and } SS_E \text{ are independent random variables}$$

Mean sums of squares

Dividing a SS by its degrees of freedom gives a **Mean Sum of Squares** (MS):

$$MS_R = \frac{SS_R}{1} \quad MS_E = \frac{SS_E}{n - 2}$$

$$\text{Note: } E[MS_E] = \frac{1}{n - 2} E[SS_E] = \sigma^2$$

So we can use $\hat{\sigma}^2 = MS_E$ as unbiased estimate of the **true variance** σ^2

Inference on regression coefficients

First, we can also prove that:

If the **true model** for the data is indeed $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, $\varepsilon_i \sim \mathbf{N}(0, \sigma^2)$ then:

- ▶ $\hat{\beta}_0$ and $\hat{\beta}_1$ are **unbiased** estimators of β_0 and β_1 respectively:

$$\mathbb{E}[\hat{\beta}_0] = \beta_0 \text{ and } \mathbb{E}[\hat{\beta}_1] = \beta_1$$

which are **jointly normal** with

$$\text{Var}[\hat{\beta}_0] = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right), \quad \text{Var}[\hat{\beta}_1] = \frac{\sigma^2}{S_{xx}}, \quad \text{Cov}[\hat{\beta}_0, \hat{\beta}_1] = -\frac{\bar{x}\sigma^2}{S_{xx}}$$

- ▶ $\hat{\beta}_0$ and $\hat{\beta}_1$ are both **independent** of SS_E

- ▶ if $\beta_1 = 0$ in the true model, then $\frac{SS_R}{\sigma^2} = \frac{MS_R}{\sigma^2} \sim \mathbf{Chi}_1^2$

→ This allows to test hypotheses on the true coefficients β_0 and β_1

Inference on regression coefficients: slope

Hypothesis $H_0: \beta_1 = \beta_{1,0}$ $\beta_{1,0}$ is the 'reference value'

$$H_A: \beta_1 \begin{matrix} > \\ \neq \\ < \end{matrix} \beta_{1,0}$$

Construction of a suitable pivot

Since $\hat{\beta}_1 \sim N(\beta_1, \frac{\sigma^2}{S_{xx}})$, we have $\frac{\hat{\beta}_1 - \beta_1}{\sigma/\sqrt{S_{xx}}} \sim N(0, 1)$

Also, $\frac{(n-2)}{\sigma^2} MS_E \sim \text{Chi}_{n-2}^2$, independently from $\hat{\beta}_1$

So, by definition of the t -distribution:
$$\frac{\frac{\hat{\beta}_1 - \beta_1}{\sigma/\sqrt{S_{xx}}}}{\sqrt{\frac{\frac{(n-2)}{\sigma^2} MS_E}{n-2}}} = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{MS_E/S_{xx}}} \sim t_{n-2}$$

Test statistic under H_0 we have $\beta_1 = \beta_{1,0}$, and the proper test statistic

$$\frac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{MS_E/S_{xx}}} \sim t_{n-2} \quad \text{under } H_0$$

Inference on regression coefficients: intercept

Hypothesis $H_0: \beta_0 = \beta_{0,0}$ $\beta_{0,0}$ is the 'reference value'

$H_A: \beta_0 \begin{matrix} > \\ \neq \\ < \end{matrix} \beta_{0,0}$

Construction of a suitable pivot

Since $\hat{\beta}_0 \sim N(\beta_0, \sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}))$, we have $\frac{\hat{\beta}_0 - \beta_0}{\sigma \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}} \sim N(0, 1)$

So, $\frac{\frac{\hat{\beta}_0 - \beta_0}{\sigma \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}}}{\sqrt{\frac{\frac{(n-2)}{\sigma^2} MS_E}{n-2}}} = \frac{\hat{\beta}_0 - \beta_0}{\sqrt{MS_E(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}})}} \sim t_{n-2}$

Test statistic under H_0 we have $\beta_0 = \beta_{0,0}$, and the proper test statistic

$$\frac{\hat{\beta}_0 - \beta_{0,0}}{\sqrt{MS_E(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}})}} \sim t_{n-2} \quad \text{under } H_0$$

Significance of the regression

Suppose that $\beta_1 = 0$ in the true model

→ the true model is the 'trivial' model $Y = \beta_0 + \epsilon$

→ this means there is no dependence of the data on x , so our best prediction is the data mean $\hat{\beta}_0 = \bar{Y}$

We can test for this situation in two equivalent ways:

- ▶ The t -test on β_1 , see 30
- ▶ An F -test based on ANOVA

If $H_0: \beta_1 = 0$ can not be rejected, we say the regression is not significant

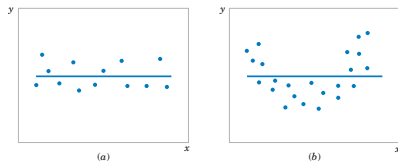


Figure 11-5 The hypothesis $H_0: \beta_1 = 0$ is not rejected.

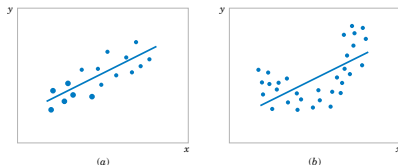


Figure 11-6 The hypothesis $H_0: \beta_1 = 0$ is rejected.

Significance of the regression: F -test

Hypothesis $H_0: \beta_1 = 0$ (i.e. regression is not useful)
 $H_A: \beta_1 \begin{matrix} > \\ \neq \\ < \end{matrix} 0$

Test statistic

We know $\frac{SS_E}{\sigma^2} = \frac{(n-2)MS_E}{\sigma^2} \sim \text{Chi}_{n-2}^2$, independent from MS_R see 28

Under H_0 we have $\beta_1 = 0$ and so $\frac{MS_R}{\sigma^2} \sim \text{Chi}_1^2$ see 29

By the definition of the F -distribution, a proper test statistic is therefore

$$\frac{\frac{MS_R}{\sigma^2 \cdot 1}}{\frac{(n-2)MS_E}{\sigma^2(n-2)}} = \frac{MS_R}{MS_E} \sim F_{1,n-2} \quad \text{under } H_0$$

Table 11-3 Analysis of Variance for Testing Significance of Regression

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Regression	$SS_R = \hat{\beta}_1 S_{xy}$	1	MS_R	MS_R/MS_E
Error	$SS_E = SS_T - \hat{\beta}_1 S_{xy}$	$n - 2$	MS_E	
Total	SS_T	$n - 1$		

Simple linear regression: confidence intervals

Confidence intervals for slope and intercept

From the pivots on 30 and 31 respectively:

A $(1 - \alpha)$ -confidence interval for the slope β_1 is

$$\hat{\beta}_1 + t_{\frac{\alpha}{2}, n-2} \sqrt{\frac{MS_E}{S_{xx}}} < \beta_1 < \hat{\beta}_1 + t_{1-\frac{\alpha}{2}, n-2} \sqrt{\frac{MS_E}{S_{xx}}}$$

A $(1 - \alpha)$ -confidence interval for the intercept β_0 is

$$\hat{\beta}_0 + t_{\frac{\alpha}{2}, n-2} \sqrt{MS_E \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)} < \beta_0 < \hat{\beta}_0 + t_{1-\frac{\alpha}{2}, n-2} \sqrt{MS_E \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)}$$

Simple linear regression: confidence intervals

Confidence intervals for the mean response at $x = x_0$

Consider a **new** or **available** data point, and give it index $i = 0$: (x_0, Y_0)

The **mean response** $\mu_{Y|x_0} = \beta_0 + \beta_1 x_0$ in this point will be estimated as

$$\hat{Y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$$

which is a linear combination of $\hat{\beta}_0$ and $\hat{\beta}_1$ and therefore also normal: see 29

$$E[\hat{Y}_0] = E[\hat{\beta}_0] + E[\hat{\beta}_1]x_0 = \beta_0 + \beta_1 x_0 = \mu_{Y|x_0}$$

$$\begin{aligned}\text{Var}[\hat{Y}_0] &= \text{Var}[\hat{\beta}_0] + 2x_0\text{Cov}[\hat{\beta}_0, \hat{\beta}_1] + x_0^2\text{Var}[\hat{\beta}_1] \\ &= \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right) + 2x_0\frac{-\bar{x}\sigma^2}{S_{xx}} + x_0^2\frac{\sigma^2}{S_{xx}} = \sigma^2\left(\frac{1}{n} + \frac{(\bar{x} - x_0)^2}{S_{xx}}\right)\end{aligned}$$

So we conclude $\hat{Y}_0 \sim \mathbf{N}(\mu_{Y|x_0}, \sigma^2(\frac{1}{n} + \frac{(\bar{x} - x_0)^2}{S_{xx}}))$

→ or after standardisation we get the pivot $\frac{\hat{Y}_0 - \mu_{Y|x_0}}{\sigma\sqrt{\frac{1}{n} + \frac{(\bar{x} - x_0)^2}{S_{xx}}}} \sim \mathbf{N}(0, 1)$

Simple linear regression: confidence intervals

Confidence intervals for the mean response at $x = x_0$

→ we can directly use this pivot for inference on $\mu_{Y|x_0}$ if σ^2 is known.

If not, use $\hat{\sigma}^2 = MS_E$ to get another pivot: $(\frac{n-2}{\sigma^2} MS_E \sim \text{Chi}^2_{n-2})$

$$\frac{\frac{\hat{Y}_0 - \mu_{Y|x_0}}{\sigma \sqrt{\frac{1}{n} + \frac{(\bar{x} - x_0)^2}{S_{xx}}}}}{\sqrt{\frac{\frac{n-2}{\sigma^2} MS_E}{n-2}}} = \frac{\hat{Y}_0 - \mu_{Y|x_0}}{\sqrt{MS_E \left(\frac{1}{n} + \frac{(\bar{x} - x_0)^2}{S_{xx}} \right)}} = \frac{\hat{Y}_0 - \mu_{Y|x_0}}{\text{se}(\hat{Y}_0)} \sim t_{n-2}$$

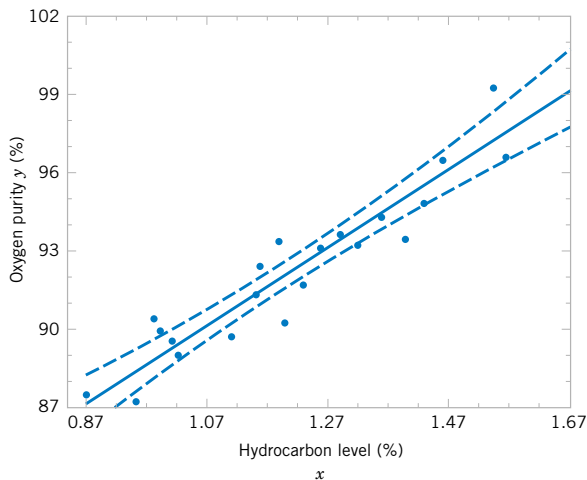
Therefore, a $(1 - \alpha)$ -confidence interval on the mean response is

$$\hat{Y}_0 + t_{\alpha/2, n-2} \text{se}(\hat{Y}_0) \leq \mu_{Y|x_0} \leq \hat{Y}_0 + t_{1-\alpha/2, n-2} \text{se}(\hat{Y}_0)$$

Simple linear regression: confidence intervals

Confidence intervals for mean response at $x = x_0$

Figure 11-7 Scatter diagram of oxygen purity data from Example 11-1 with fitted regression line and 95 percent confidence limits on $\mu_{Y|x_0}$.



Simple linear regression: prediction intervals

Prediction interval on a future observation at $x = x_0$

Consider again a **new** unobserved data point (x_0, Y_0)

then we **predict** the response Y_0 by $\hat{Y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$ (this is the best we can do)

→ prediction error is $\hat{Y}_0 - Y_0 = -e_0 = \hat{Y}_0 - \mu_{Y|x_0} - \varepsilon_0$

which is again normal with,

see 35 and 28

$$E[e_0] = E[\hat{Y}_0] - \mu_{Y|x_0} = 0$$

$$\text{Var}[e_0] = \text{Var}[\hat{Y}_0] + \underbrace{\text{Var}[\varepsilon_0]}_{\sigma^2} = \sigma^2 \left(1 + \frac{1}{n} + \frac{(\bar{x} - x_0)^2}{S_{xx}} \right)$$

So similar as on 36 we have

$$\frac{e_0 - 0}{\text{se}(e_0)} = \frac{Y_0 - \hat{Y}_0}{\text{se}(e_0)} \sim t_{n-2} \quad \text{with} \quad \text{se}(e_0) = \sqrt{\text{MSE} \left(1 + \frac{1}{n} + \frac{(\bar{x} - x_0)^2}{S_{xx}} \right)}$$

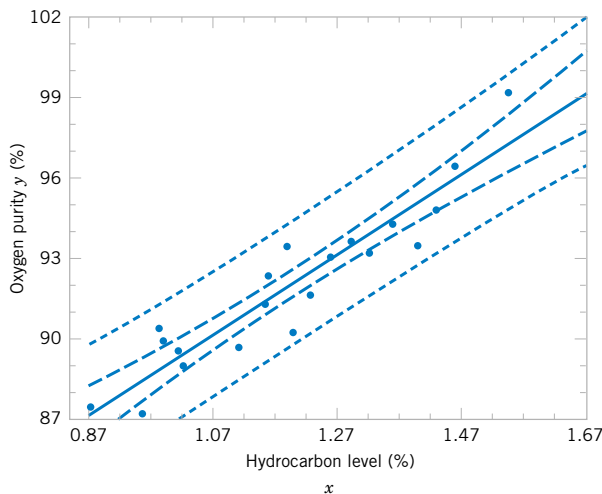
Therefore a $(1 - \alpha)$ -**prediction interval** on a future observation Y_0 at x_0 is

$$\hat{Y}_0 + t_{\alpha/2, n-2} \text{se}(e_0) \leq Y_0 \leq \hat{Y}_0 + t_{1-\alpha/2, n-2} \text{se}(e_0)$$

Simple linear regression: prediction intervals

Prediction interval on a future observation at $x = x_0$

Figure 11-8 Scatter diagram of oxygen purity data from Example 11-1 with fitted regression line, 95% prediction limits (outer lines) and 95% confidence limits on $\mu_{Y|x_0}$.



Checking model adequacy

Recall 16 we assumed a true model for the data points as follows:

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \text{ with } \varepsilon_i \text{ iid } \mathcal{N}(0, \sigma^2), i = 1, \dots, n$$

So the assumptions we made basically are:

- ▶ The model is linear in x
- ▶ The errors ε_i are independent of each other
- ▶ The errors ε_i are all normal with zero mean and the same variance

This model is not necessarily appropriate for all possible data sets (x_i, y_i) !

→ The analyst should always consider the validity of these assumptions to be doubtful and conduct analysis to examine the adequacy of the model

Post hoc analysis

How to check the validity of a model once it is fitted and found to be significant?

- ▶ We can do this graphically, by plotting the data.
although this is difficult for higher dimensional x -space, see later
- ▶ We can use the residuals e_i as a proxy for the errors ε_i
various kinds of residuals plots, Q-Q plot, formal tests, ...

Checking model adequacy

Post hoc analysis of residuals

Usually the **residuals** are 'close' to the errors although theoretically, (see later)

- they are slightly correlated
- they are normal with mean 0 but have slightly different variances

Nevertheless, we can make different kinds of **residuals plots** to check visually:

- ▶ Do the residuals have **constant variance**?
- ▶ Are the residuals **normally** distributed?

If not, **the model might not be adequate** and a better one (with other x -variable) should be investigated.

Possible residuals plots are for example

- ▶ Residuals vs. fitted ($\hat{Y}_i, e_i$), vs. response (y_i, e_i) or vs. values (x_i, e_i)
there should be no obvious patterns: variability $\pm$ same everywhere
- ▶ Scale-location plot ($\hat{Y}_i, e_i/\sqrt{\hat{\sigma}^2}$) most should be within $(-2, 2)$
- ▶ Normal probability plot of the residuals points close to diagonal
- ▶ Residuals vs. leverages ($h_{ii}, e_i/\sqrt{\hat{\sigma}^2}$) (see later)

Checking model adequacy

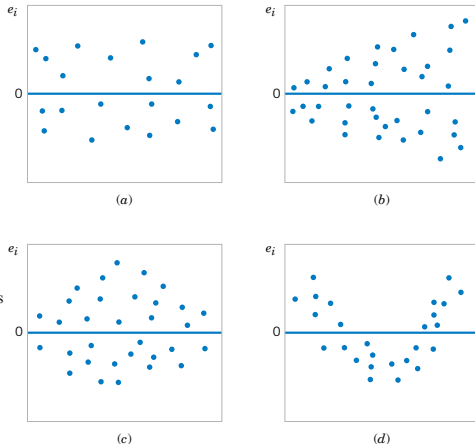


Figure 11-9 Patterns for residual plots.
(a) satisfactory, (b) funnel, (c) double bow, (d) nonlinear.
[Adapted from Montgomery, Peck, and Vining (2001).]

Such patterns indicate that a better model may be found by

- ▶ using additional or other x -variables to capture nonlinear relationships
- ▶ using a **transformed** response ($\sqrt{y}$, $\ln y$, $1/y$, ...) to stabilise the variance

Checking model adequacy

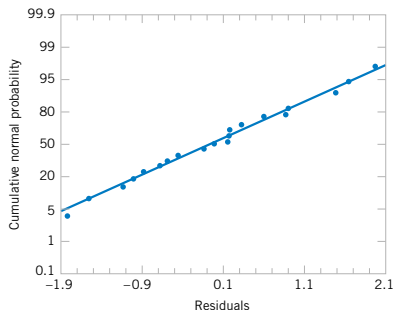


Figure 11-10 Normal probability plot of residuals, Example 11-7.

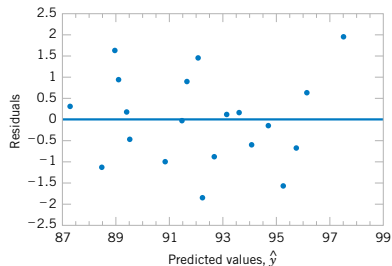


Figure 11-11 Plot of residuals versus predicted oxygen purity $\hat{y}$, Example 11-7.

Checking model adequacy

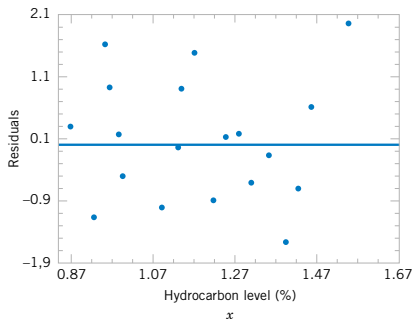


Figure 11-12 Plot of residuals versus hydrocarbon level x , Example 11-8.

Effect of outliers

The effect of a single outlier on the regression line

- ▶ may be fairly modest if it is close to mean $\bar{x}$ in x -space (affects intercept a bit)
- ▶ may be quite severe when at either extreme of x -range (affects the slope)
→ such outlier is an influential observation

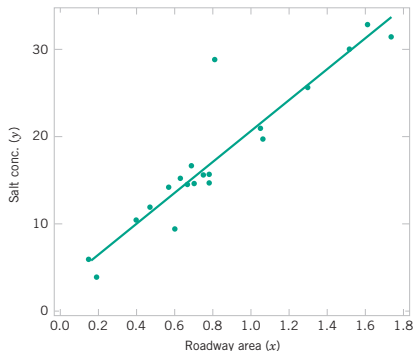


Figure 6-15 Effect of an outlier.

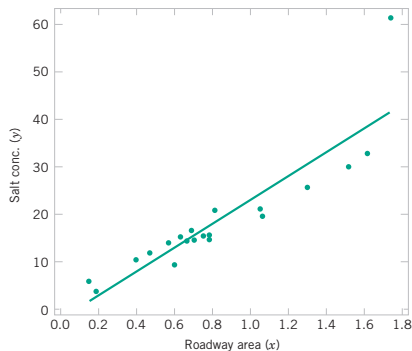


Figure 6-16 Effect of an influential observation.

MEERVOUDIGE LINEAIRE REGRESSIE: MODEL

The diagram shows the linear regression equation $Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i$ with the following annotations:

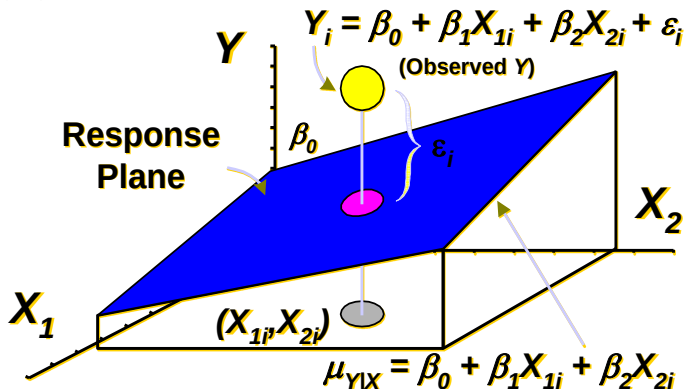
- respons**: points to Y_i
- intercept**: points to β_0
- richtings-coëfficiënten**: points to the slope coefficients $\beta_1, \beta_2, \dots, \beta_k$
- foutterm**: points to the error term ε_i
- A blue bracket under the term $\beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki}$ is labeled **waarden x-variabelen bij observatie i**
- The index $i = 1, \dots, n$ is shown at the bottom right, with an arrow pointing to it from the text **aantal metingen**.
- A note at the bottom left states: **Alle fouten ε_i zijn normaal verdeeld**

Alle fouten ε_i zijn normaal verdeeld
rond 0 en met standaarddeviatie σ

→ $k + 2$ vrije parameters om te schatten!

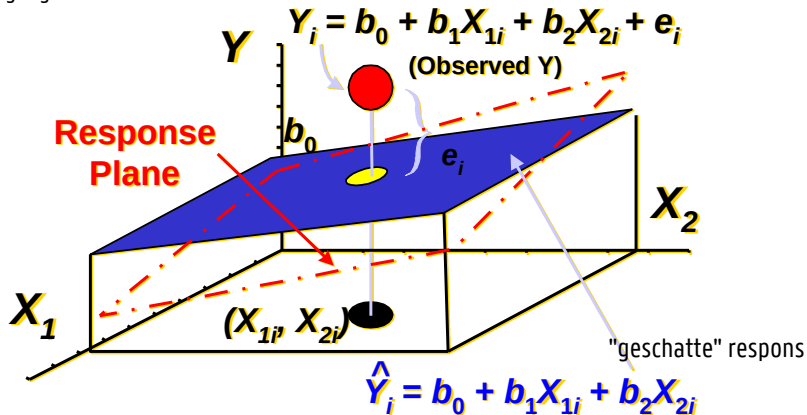
MEERVOUDIGE LINEAIRE REGRESSIE: MODEL (POPULATIE)

tweevoudig regressiemodel

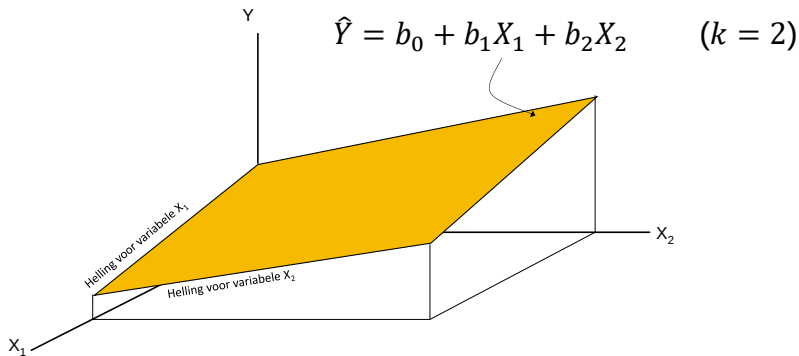


MEERVOUDIGE LINEAIRE REGRESSIE: GESCHATTE RESPONS

tweevoudig regressiemodel



MEERVOUDIGE LINEAIRE REGRESSIE: GESCHATTE RESPONS



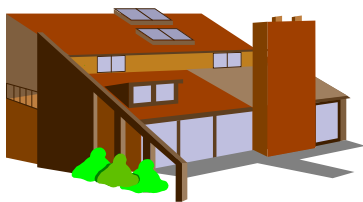
ENKELVOUDIGE EN MEERVOUDIGE REGRESSIE VERGELEKEN

- Coëfficiënt β_1 van variabele X in een **enkelvoudig** regressiemodel omvat de impact van X maar ook van **andere** variabelen die gecorreleerd zijn met X en de respons Y
→ verwarring, confounding
- De coëfficiënten van de variabelen $X_1, X_2, \dots, X_k$ in een **meervoudig** regressiemodel zullen maximaal de impact van deze (gecorrleerde) variabelen onderling onderscheiden.

Er is dus zeker een voordeel aan het gezamenlijk modeleren van alle variabelen in één regressiemodel om de respons te verklaren (en voor nieuwe metingen: te voorspellen).

MEERVOUDIGE LINEAIRE REGRESSIE: VOORBEELD

Bouw een model om het olieverbruik (in gallons) te schatten voor de verwarming van een ééngesinswoning in de maand januari gebaseerd op gemiddelde maandtemperatuur (in °F) en dikte van de isolatie (in inches).



Beschikbare data voor 15 huizen:

Oil (Gal)	Temp	Insulation
275,30	40	3
363,80	27	3
164,30	40	10
40,80	73	6
94,30	64	6
230,90	34	6
366,70	9	6
300,60	8	10
237,80	23	10
121,40	63	3
31,40	65	10
203,50	41	6
441,10	21	3
323,00	38	3
52,50	58	10

MEERVOUDIGE LINEAIRE REGRESSIE: VOORBEELD

volledige Excel output:



Regression Statistics	
Multiple R	0.98265476
R Square	0.96561037
Adjusted R Square	0.95987877
Standard Error	26.0137832
Observations	15

Dit noemen we **MSE**, zie verder

ANOVA					
	df	SS	MS	F	Significance F
Regression	2	228014.6263	114007.3	168.471203	1.65411E-09
Residual	12	8120.603016	676.7169		
Total	14	236135.2293			

geschatte coëfficiënten
 $b_0, b_1, \dots$ in de
regressievergelijking

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	562.151009	21.09310433	26.65094	4.7787E-12	516.1930829	608.1089356
Temp	-5.4365806	0.336216167	-16.1699	1.6418E-09	-6.16913269	-4.704028491
Insulation	-20.012321	2.342505227	-8.54313	1.9073E-06	-25.1162011	-14.90844022

MEERVOUDIGE LINEAIRE REGRESSIE: VOORBEELD

Excel output: geschatte coëfficiënten

	<i>Coefficients</i>
Intercept	562.151009
Temp	-5.4365806
Insulation	-20.012321



Denk steeds na over het aantal beduidende cijfers dat je rapporteert. Excel trekt zich daar niks van aan, maar een ingenieur wél.


$$\hat{Y} = 562.15 - 5.44X_1 - 20.01X_2$$

Voor een toename van de temperatuur met 1°F daalt het geschatte olieverbbruik met 5.44 gallons, bij gelijke isolatiedikte.

Voor een toename van de isolatiedikte met 1 inch daalt het geschatte olieverbbruik met 20.01 gallons, bij gelijke temperatuur.



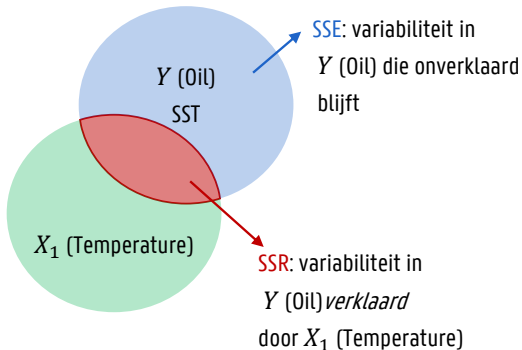
MEERVOUDIGE LINEAIRE REGRESSIE: VERKLARENDE KRACHT

De kwadratensom van een model (**SSR**) is een maat voor de verklarende kracht van dat model.

R^2 geeft de fractie van de totale variabiliteit SST die wordt verklaard door het model (dus door de variabelen in dat model)

$$R^2 = \frac{\text{SSR}}{\text{SST}}$$

enkelvoudig model $Y = \beta_0 + \beta_1 X_1 + \varepsilon$



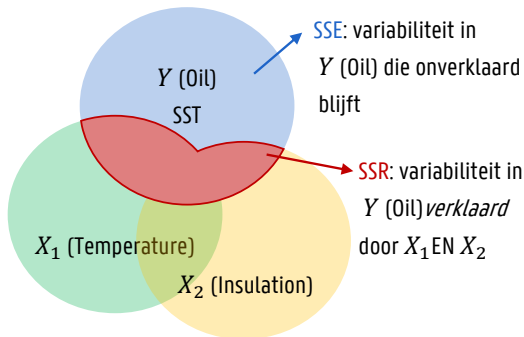
MEERVOUDIGE LINEAIRE REGRESSIE: VERKLARENDE KRACHT

De kwadratensom van een model (**SSR**) is een maat voor de verklarende kracht van dat model.

Als we een model groter maken door er een **variabele aan toe te voegen**, zal ook zijn **SSR** toenemen en dus een groter deel van de totale variabiliteit van de respons (SST) verklaren.

$$\rightarrow R^2 = \frac{SSR}{SST} \text{ stijgt}$$

$$\text{meervoudig model } Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$



MEERVOUDIGE LINEAIRE REGRESSIE: VERKLARENDE KRACHT

- Op die manier: hoe groter het model (hoe meer variabelen), hoe groter R^2 .
Maar de toegevoegde variabelen zijn **niet altijd significant**: hun bijdrage aan **SSR** is relatief klein en we zijn niet erg zeker of de bijhorende coëfficiënten wel écht verschillen van 0.
- Daarom gebruiken we liever de **aangepaste R^2** als maat van hoe goed een model is:

$$R_{\text{adj}}^2 = 1 - (1 - R^2) \frac{n-1}{n-k-2}$$

Deze R_{adj}^2 is wél een goed maat: zal alleen stijgen bij het toevoegen van een factor die voldoende significant is.

- R_{adj}^2 is dus nuttig om verschillende modellen te vergelijken: **hoe hoger, hoe beter**

MEERVOUDIGE REGRESSIE: VOORSPELLINGEN

Wat is nu het nut van zo'n regressiemodel? Wat kunnen we ermee doen?

1. Het model geeft een **voorspelling** van de **respons bij een nieuw datapunt**



Bijvoorbeeld: schat het oliegebruik
voor een ander huis wanneer de januari-
temperatuur **30°F** graden is en de isolatie
6 inches dik

$$\begin{aligned}\hat{Y} &= 562.15 - 5.43 \cdot 30 - 20.01 \cdot 6 \\ &= 278.97\end{aligned}$$

Het **voorspelde** oliegebruik voor dat huis is 278.97 gallons!

2. Het model geeft ook een schatting van σ :

$$\hat{\sigma} = \sqrt{\text{MSE}} = \sqrt{\frac{\text{SSE}}{n - k - 1}} = \sqrt{\frac{8120.6}{15 - 2 - 1}} = 26.0$$

Deze $\hat{\sigma}$ kunnen we zien als de standaarddeviatie van zo'n voorspelling
(min of meer), dus hoeveel onzekerheid op de voorspelling zit.

Other Aspects of Regression

Polynomial Models

In Section 6-1 we observed that models with polynomial terms in the regressors, such as the second-order model

$$Y = \beta_0 + \beta_1 x_1 + \beta_{11} x_1^2 + \epsilon$$

are really linear regression models and can be fit and analyzed using the methods discussed in Section 6-3. Polynomial models arise frequently in engineering and the sciences, and this contributes greatly to the widespread use of linear regression in these fields.

Table 6-9 The Acetylene Data

Observation	Yield, Y	Temp, T	Ratio, R	Observation	Yield, Y	Temp, T	Ratio, R
1	49.0	1300	7.5	9	34.5	1200	11.0
2	50.2	1300	9.0	10	35.0	1200	13.5
3	50.5	1300	11.0	11	38.0	1200	17.0
4	48.5	1300	13.5	12	38.5	1200	23.0
5	47.5	1300	17.0	13	15.0	1100	5.3
6	44.5	1300	23.0	14	17.0	1100	7.5
7	28.0	1200	5.3	15	20.5	1100	11.0
8	31.5	1200	7.5	16	29.5	1100	17.0

Other Aspects of Regression

Polynomial Models

The Minitab regression output for this reduced model is as follows.

The regression equation is

$\text{Yield} = 36.1 + 0.134 \text{ Temp} + 0.351 \text{ Ratio}$

Predictor	Coef	SE Coef	T	P	VIF
Constant	36.1063	0.9060	39.85	0.000	
Temp	0.13396	0.01191	11.25	0.000	1.1
Ratio	0.3511	0.1696	2.07	0.059	1.1

S = 3.624

R-Sq = 92.0%

R-Sq(adj) = 90.7%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	2	1952.98	976.49	74.35	0.000
Residual Error	13	170.73	13.13		
Total	15	2123.71			

Other Aspects of Regression

Polynomial Models

The second-order model in two regressors is

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \epsilon$$

$$Y = \beta_0 + \beta_1(T - 1212.5) + \beta_2(R - 12.444) \\ + \beta_{12}(T - 1212.5)(R - 12.444) + \beta_{11}(T - 1212.5)^2 + \beta_{22}(R - 12.444)^2 + \epsilon$$

Other Aspects of Regression

Polynomial Models

$$H_0: \beta_{r+1} = \beta_{r+2} = \cdots = \beta_k = 0$$

$$H_1: \text{At least one of the } \beta\text{'s} \neq 0$$

The test statistic for the above hypotheses was originally given in equation 6-56, repeated below for convenience:

$$F_0 = \frac{[SS_E(RM) - SS_E(FM)]/(k - r)}{SS_E(FM)/(n - p)}$$

Other Aspects of Regression

Polynomial Models

The regression equation is

$$\text{Yield} = 36.4 + 0.130 \text{ Temp} + 0.480 \text{ Ratio} - 0.00733 \text{ T} \times \text{R} + 0.000178 \text{ T}^2 - 0.0237 \text{ R}^2$$

Predictor	Coef	SE Coef	T	P	VIF
Constant	36.4339	0.5529	65.90	0.000	
Temp	0.130476	0.003642	35.83	0.000	1.1
Ratio	0.48005	0.05860	8.19	0.000	1.5
T × R	-0.0073346	0.0007993	-9.18	0.000	1.4
T^2	0.00017820	0.00005854	3.04	0.012	1.2
R^2	-0.02367	0.01019	-2.32	0.043	1.7

S = 1.066

R-Sq = 99.5%

R-Sq (adj) = 99.2%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	5	2112.34	422.47	371.49	0.000
Residual Error	10	11.37	1.14		
Total	15	2123.71			

Other Aspects of Regression

Variable Selection Procedures

Best Subsets Regressions

Table 6-10 Minitab Best Subjects Regression for Shampoo Data

Response is Quality

Vars	R-Sq	R-Sq(adj)	Mallows C-p	S	R e s e n s i b i l i t y				
					1	2	3	4	5
1	26.2	22.9	46.4	3.8321	X				
1	25.7	22.3	46.9	3.8455					
1	23.9	20.5	48.5	3.8915					
1	6.3	2.1	64.3	4.3184	X				
1	3.8	0.0	66.7	4.3773		X			
2	62.2	58.6	16.1	2.8088	X	X			
2	50.3	45.6	26.7	3.2185	X		X		
2	42.6	37.2	33.6	3.4589			X	X	
2	40.9	35.3	35.2	3.5098	X	X			
2	32.6	26.2	42.7	3.7486	X	X			
3	77.6	74.2	4.2	2.2164	X		X	X	
3	63.1	57.6	17.2	2.8411	X	X	X		
3	62.5	56.9	17.7	2.8641	X	X	X		
3	52.9	45.9	26.4	3.2107	X	X	X		
3	51.8	44.6	27.4	3.2491	X	X		X	
4	79.9	75.7	4.1	2.1532	X	X	X	X	
4	78.6	74.1	5.3	2.2205	X	X	X	X	
4	64.8	57.4	17.7	2.8487	X	X	X	X	
4	53.0	43.1	28.3	3.2907	X	X	X	X	
4	51.4	41.2	29.7	3.3460		X	X	X	X
5	80.0	74.5	6.0	2.2056	X	X	X	X	X

Other Aspects of Regression

Variable Selection Procedures

Backward Elimination

Table 6-11 Stepwise Regression Backward Elimination for Shampoo
Data: Quality versus Foam, Scent, Color, Residue, Region

Backward elimination. Alpha-to-Remove: 0.1			
Response is Quality on 5 predictors, with N = 24			
Step	1	2	3
Constant	86.33	86.14	89.81
Foam	1.82	1.87	1.82
T-Value	5.07	5.86	5.58
P-Value	0.000	0.000	0.000
Scent	1.03	1.18	
T-Value	1.12	1.48	
P-Value	0.277	0.155	
Color	0.23		
T-Value	0.33		
P-Value	0.746		
Residue	-4.00	-3.93	-3.38
T-Value	-4.93	-5.15	-4.93
P-Value	0.000	0.000	0.000
Region	-3.86	-3.71	-3.41
T-Value	-3.70	-4.05	-3.70
P-Value	0.002	0.001	0.001
S	2.21	2.15	2.22
R-Sq	80.01	79.89	77.57
R-Sq (adj)	74.45	75.65	74.20
Mallows C-p	6.0	4.1	4.2

Other Aspects of Regression

Variable Selection Procedures

Forward Selection

Table 6-12 Stepwise Regression Forward Selection for Shampoo Data:
Quality versus Foam, Scent, Color, Residue, Region

Forward selection. Alpha-to-Enter: 0.25

Response is Quality on 5 predictors, with N = 24

Step	1	2	3	4
Constant	76.00	89.45	89.81	86.14
Foam	1.54	1.90	1.82	1.87
T-Value	2.80	4.61	5.58	5.86
P-Value	0.010	0.000	0.000	0.000
Residue		-3.82	-3.38	-3.93
T-Value		-4.47	-4.93	-5.15
P-Value		0.000	0.000	0.000
Region			-3.41	-3.71
T-Value			-3.70	-4.05
P-Value			0.001	0.001
Scent				1.18
T-Value				1.48
P-Value				0.155
S	3.83	2.81	2.22	2.15
R-Sq	26.24	62.17	77.57	79.89
R-Sq (adj)	22.89	58.57	74.20	75.65
Mallows C-p	46.4	16.1	4.2	4.1

Other Aspects of Regression

Variable Selection Procedures

Stepwise Regression

Table 6-13 Stepwise Regression Combined Forward and Backward Elimination: Quality versus Foam, Scent, Color, Residue, Region

Alpha-to-Enter: 0.15 Alpha-to-Remove: 0.15			
Response is Quality on 5 predictors, with N = 24			
Step	1	2	3
Constant	76.00	89.45	89.81
Foam	1.54	1.90	1.82
T-Value	2.80	4.61	5.58
P-Value	0.010	0.000	0.000
Residue		-3.82	-3.38
T-Value		-4.47	-4.93
P-Value		0.000	0.000
Region			-3.41
T-Value			-3.70
P-Value			0.001
S	3.83	2.81	2.22
R-Sq	26.24	62.17	77.57
R-Sq (adj)	22.89	58.57	74.20
Mallows C-p	46.4	16.1	4.2

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